

Sampathkumar N

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Introduction

Final-year B.E. student in Computer Science and Engineering (AI & ML) with a strong interest in machine learning, computer vision, and retrieval-augmented generation. Experienced in designing, building, and evaluating deep learning and LLM-based systems through research projects, patent work, and industry internships. Seeking a graduate research position to deepen my foundation in machine learning and contribute to rigorous, well-evaluated research.

Education

Easwari Engineering College (SRM Group)
B.E., Computer Science and Engineering (Artificial Intelligence & Machine Learning)
• CGPA: 8.82 / 10

Chennai, India
Aug 2023 – May 2027

Patents & Publications

All patents filed through Easwari Engineering College (SRM Group).

Research paper accepted for presentation at ARIIA-2026 — Second International Conference on Augmented Reality, Intelligent Systems and Industrial Automation (Dec 2026); proceedings published by Taylor & Francis.

Knowledge Dark Matter Retrieval Network (KDM-RN) — Patent Filed. Framework for inferring latent knowledge absent from existing knowledge graphs. (Co-inventors: Duranjai V, Ilamurugan G, Deeraj Kumar M.)

GeoMemory+: Terrain-Constrained Reconstruction of Historical Geography — Patent Filed. Diffusion model reconstructing historical terrain from satellite and DEM data. (Co-inventors: Duranjai V, Ilamurugan G.)

EduPulse AI: Personalized Learning & Career Guidance System — Patent Filed. Multilingual adaptive learning platform with an adaptive scoring engine. (Co-inventors: Duranjai V, Ilamurugan G.)

BaristaPOV VLM: Automated Analysis of Barista POV Video Streams — Patent Filed. Vision-language framework scoring beverage-preparation workflows. (Co-inventors: Duranjai V, Ilamurugan G.)

Research Projects

AutoPatch — Agentic AST-Based CI/CD Debugger and Autonomous Patcher
Python, AST, FastAPI
• Localizes test failures via AST traversal and synthesizes verified patch diffs — 100% pass@1 in 0.15s per fix.

HyperVector — Sub-Millisecond SIMD Vector Search Engine
C++20, ARM NEON, HNSW
• HNSW vector index with hand-written 128-bit NEON intrinsics — 21,750+ QPS at 0.046ms latency, 99.4% recall@10.

EdgeDepth-MPS — Real-Time Monocular-to-3D Vision on Apple Metal
PyTorch, MPS, Point Clouds
• Depth estimation with camera-intrinsic unprojection to dense point clouds — 46.8 FPS on-device.

EchoStream — Full-Duplex On-Device Voice AI
Apple MLX, WebSockets, VAD
• Streaming voice pipeline with barge-in handling — 23.66ms glass-to-glass latency, <40ms interrupt response.

NeuroScan-3D — Volumetric 3D MRI Brain Tumor Segmentation
PyTorch, 3D-UNet, BraTS
• Multi-modal 3D-UNet with multi-planar reconstruction — 94.1% Dice on whole tumor, 92.4% on tumor core.

NeuroBlink — Neuromorphic SNN for Drone Collision Evasion
SNN, Event Cameras, LIF Neurons
• Spiking neural network on event-camera data — 2.88ms evasion latency, 1,000x finer temporal resolution than frame cameras.

Experience

Machine Learning Intern, Tendworks Pvt Ltd
Chennai, India
• Developed data-handling and workflow logic for the HRMS Expense Management module.

Mar 2026 – Sep 2026

Machine Learning Intern, OEG Defence & Aerospace Pvt. Ltd.
Chennai, India
• Built and deployed YOLOv8 drone object-detection models with Roboflow datasets and OpenCV inference pipelines.

Jun 2025

Technical Skills

Languages: Python, C++, C, Java, SQL, JavaScript/TypeScript
AI/ML: PyTorch, Computer Vision, LLMs, RAG, Vision-Language Models, Spiking Neural Networks, Diffusion Models
Systems & Tools: Git, Docker, FastAPI, OpenCV, Roboflow, Vector Databases (pgvector, Redis), SQLite

Honors & Activities

Participant, Smart India Hackathon 2024 (RailNav railway navigation system).
Participant: Convolve 4.0 (Pan-IIT AI/ML Hackathon), Dev Heat Hackathon (Spring Fiesta), Hack-4-Mini 2.0.

Certifications

Cisco Networking Academy: CCNA — Introduction to Networks; Switching, Routing & Wireless Essentials; Enterprise Networking, Security & Automation; Network Technician Career Path; Python Essentials 1 & 2; Introduction to Modern AI; Data Analytics Essentials
Oracle: OCI 2025 Certified Generative AI Professional; OCI 2025 AI Foundations Associate
NPTEL: Internet of Things; Privacy & Security in Online Social Media; Leadership & Team Effectiveness